

MATH 4181H: Induction and Well-Ordering

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1 A Positivity Exercise

Theorem 1. *Let $x, y \in \mathbb{R}$, not both zero. Then*

$$x^4 + x^2y + y^2 > 0.$$

Proof. Complete the square:

$$x^4 + x^2y + y^2 = \left(x^2 + \frac{y}{2}\right)^2 + \frac{3}{4}y^2.$$

Both terms on the right are nonnegative. If their sum were zero, then $y = 0$ and $x^2 + y/2 = 0$, which would also give $x = 0$. This contradicts the assumption that x and y are not both zero. Therefore the sum is strictly positive.

The factorization appearing in the handwritten argument is

$$x^6 - y^3 = (x^2 - y)(x^4 + x^2y + y^2).$$

It gives another proof: if $x^2 \neq y$, then $x^6 - y^3$ and $x^2 - y$ have the same sign because the cube function is strictly increasing, so their quotient is positive. If $x^2 = y$, then

$$x^4 + x^2y + y^2 = 3x^4 > 0,$$

since x and y cannot both be zero. □

2 Natural Numbers and Mathematical Induction

Let

$$\mathbb{N} = \{1, 2, 3, \dots\}.$$

One way to characterize the natural numbers axiomatically is through Peano arithmetic. A central principle is mathematical induction.

Principle 1 (Mathematical induction). *Let $P(k)$ be a statement that is either true or false for each $k \in \mathbb{N}$. Suppose that*

- (1) $P(1)$ is true;
 (2) for every $k \in \mathbb{N}$, if $P(k)$ is true, then $P(k+1)$ is true.

Then $P(k)$ is true for every $k \in \mathbb{N}$.

Example: the sum of the first k natural numbers

We prove that, for every $k \in \mathbb{N}$,

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}.$$

Proof. Let $P(k)$ denote the displayed identity.

Base case. For $k = 1$,

$$1 = \frac{1(1+1)}{2},$$

so $P(1)$ is true.

Inductive step. Assume $P(k)$ is true; that is,

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}.$$

Adding $k+1$ to both sides gives

$$\begin{aligned} 1 + 2 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= (k+1) \left(\frac{k}{2} + 1 \right) \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

This is precisely $P(k+1)$. Therefore the formula holds for every $k \in \mathbb{N}$ by induction. □

3 The Well-Ordering Principle

Principle 2 (Well-ordering principle). *If $A \subseteq \mathbb{N}$ and $A \neq \emptyset$, then A has a least element.*

The well-ordering principle and mathematical induction are equivalent. We prove both implications.

Well-ordering implies induction

Assume the well-ordering principle. Suppose $P(k)$ satisfies the hypotheses of mathematical induction: $P(1)$ is true, and $P(k) \Rightarrow P(k+1)$ for every $k \in \mathbb{N}$.

Suppose, for a contradiction, that $P(k)$ is false for at least one natural number. Define

$$A = \{k \in \mathbb{N} : P(k) \text{ is false}\}.$$

Then A is nonempty, so it has a least element ℓ . Because $P(1)$ is true, $\ell \neq 1$; hence $\ell > 1$ and $\ell - 1 \in \mathbb{N}$.

By the minimality of ℓ , the statement $P(\ell - 1)$ is true. The inductive implication then gives $P(\ell)$, contradicting the fact that $\ell \in A$. Thus A must be empty, and $P(k)$ is true for every $k \in \mathbb{N}$.

Induction implies well-ordering

Assume mathematical induction. Suppose, for a contradiction, that there is a nonempty set $A \subseteq \mathbb{N}$ with no least element.

For $k \in \mathbb{N}$, let $P(k)$ be the statement

$$1, 2, \dots, k \notin A.$$

Base case. We must have $1 \notin A$, because if $1 \in A$, then 1 would be the least element of A . Therefore $P(1)$ is true.

Inductive step. Assume $P(k)$ is true, so none of $1, 2, \dots, k$ lies in A . If $k + 1 \in A$, then $k + 1$ would be the least element of A , again a contradiction. Hence $k + 1 \notin A$, and $P(k + 1)$ is true.

By induction, $P(k)$ is true for every $k \in \mathbb{N}$. Thus no natural number belongs to A , which says $A = \emptyset$. This contradicts the assumption that A is nonempty. Therefore every nonempty subset of $\mathbb{N}$ has a least element.

4 Strong Induction

Principle 3 (Strong induction). *Let $P(k)$ be a statement for each $k \in \mathbb{N}$. Suppose that*

(1) $P(1)$ is true;

(2) for every $k \geq 2$, if $P(1), P(2), \dots, P(k - 1)$ are all true, then $P(k)$ is true.

Then $P(k)$ is true for every $k \in \mathbb{N}$.

Strong induction allows the proof of $P(k)$ to use all preceding cases, rather than only the immediately preceding case $P(k - 1)$.